Mississippi State University
Department of Computer Science and Engineering

Introduction to Machine Learning

NSF ExLENT XAI Initiative

Principal Investigators: Dr. Vini Chaudhary and Dr. Charan Gudla

Course Information

Modality:	Hybrid (In-Person + Online)
Location:	Butler Hall 102
Schedule:	Friday 3:30 PM - 5:00 PM (online) | Saturday 12:00 PM - 2:00 PM (In-Person)
Instructor:	Dr. Nisha Pillai
Email:	pillai@cse.msstate.edu (Response within 24 hours)
Prerequisites:	Python programming experience
Software:	Python, Scikit-learn, TensorFlow, PyTorch, and related deep learning libraries

Catalog Description

This course provides a comprehensive introduction to machine learning concepts, techniques, and applications. Students will explore the full pipeline from data preprocessing through model deployment, with a strong emphasis on ethical and explainable AI. Topics span supervised and unsupervised learning, deep learning with neural networks, responsible AI practices, and real-world applied capstone projects.

Learning Outcomes

By the end of this course, students will be able to:

- Apply machine learning concepts and tools to analyze real-world datasets, build predictive models, and evaluate their performance critically.
- Develop responsible AI practices by understanding the ethical, fairness, and privacy implications of deploying machine learning systems.
- Demonstrate technical proficiency by completing an end-to-end applied project, from data processing to model deployment, and communicating findings effectively.

Course Topics & Schedule

Phase 1 - Foundations to Deployment (Weeks 1-13)

Week(s)	Topic	Assessments
Weeks 1-2	Foundations: AI/ML overview, Python, NumPy, Pandas, and AI ethics	
Weeks 3-4	Data Processing: Feature engineering, encoding, scaling, handling missing data	

Weeks 5-6	Supervised Learning: Regression, Decision Trees, SVM, model evaluation	Assignment 1, Quiz 1
Week 7	Unsupervised Learning: Clustering, PCA, Gaussian Mixture Models	Assignment 2
Weeks 8-10	Deep Learning: Neural networks, Keras/TensorFlow, CNNs	Assignment 3
Week 11	Responsible AI: Fairness, bias, privacy, federated learning	Quiz 2
Weeks 12-13	Deployment: Docker, FastAPI, MLflow, transfer learning	Assignment 4

Phase 2 - Capstone Project (Weeks 14 - 16)

Week(s)	Topic	Assessments
Weeks 14-16	Capstone Project: Applied project with final presentation and evaluation	Final Presentation & Evaluation

Grading

Final grades will be computed based on the following weighted assignments.

Assignments (4)	30%
Quiz (2)	30%
Capstone Project	40%
Total	**100%**

Grading Scale

Grade	**Range**
A	90–100%
B	80–89%
C	70–79%
D	60–69%
F	Below 60%

Capstone Project

The capstone project is designed to assess students' ability to integrate and apply the skills developed throughout the semester to a real-world problem.

- Students will work in teams of 2 or 3 members.
- Each group will develop a complete ML pipeline from data preprocessing to model evaluation and deployment.
- Grading (40%) is divided across three components:

- Implementation & Code (20%), which evaluates technical correctness, code quality, and effective use of ML tools and frameworks.
- Written Report (10%), which assesses the depth of analysis, documentation of methodology, and interpretation of results.
- Final Presentation & Evaluation (10%), which measures the student's ability to communicate their approach, findings, and limitations clearly to an audience.

Attendance Policy

While students are responsible for keeping up with lecture content, reviewing course materials, and submitting assignments on time. Students are encouraged to communicate proactively with the instructor regarding any absences.

Generative AI Policy

Students may use tools like ChatGPT for designated assignments. When using AI-generated content, students must follow the citation guidelines, including proper attribution through in-text citations, quotations, and references. Students must also include the following statement in assignments to indicate use of a Generative AI Tool:

> *"The author acknowledges the use of [Generative AI Tool Name] in the preparation of this assignment for [brainstorming, grammatical correction, citation, etc.]."*

University Syllabus

The Mississippi State University Syllabus is the authoritative document containing all official policies and procedures for both in-person and online courses. These University policies supersede any conflicting information found elsewhere. Each student is responsible for reviewing and understanding these policies to meet University expectations.

The University Syllabus may be accessed at any time on the Provost website under Faculty and Student Resources at: https://www.provost.msstate.edu/faculty-student-resources/university-syllabus